

2.1: Radians

Alternate way of measuring angles, such that it describes the ratio of the radius to the circumference of the circle

Therefore:

$2\pi = 360^\circ$ Deg → Rad: $n^\circ \times \frac{\pi}{180}$ Rad → Deg: $n^c \times \frac{180}{\pi}$

2.2: Circular Measures

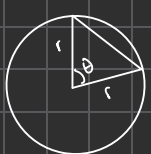
For Arc Length:

$P = 2\pi r$, Fraction: $\frac{\theta}{2\pi}$ $\Rightarrow S = 2\pi r \times \frac{\theta}{2\pi} = r\theta$

Area of a Sector:

$A = \pi r^2$, Fraction: $\frac{\theta}{2\pi}$ $\Rightarrow A_S = \pi r^2 \times \frac{\theta}{2\pi} = \frac{1}{2} r^2 \theta$

Area of a Segment:

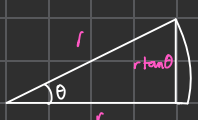


$A_{SEC} = \frac{1}{2} r^2 \theta$

$A_{TRI} = \frac{1}{2} r^2 \sin\theta$

$\Rightarrow A_{SEG} = A_{SEC} - A_{TRI} = \frac{1}{2} r^2 (\theta - \sin\theta)$

2.3: Small Angle Approximations



For small θ, the curve is approximately a straight line, so it can be seen as a triangle.

$\Rightarrow A = \frac{1}{2} r^2 \theta \approx \frac{1}{2} r^2 \sin\theta \approx \frac{1}{2} r^2 \tan^2\theta$

$\Rightarrow \theta \approx \sin\theta \approx \tan\theta$

For Cosine:

$\sin\left(\frac{\theta}{2}\right) \approx \frac{\theta}{2}$ $\cos(\theta) = 1 - 2\sin^2\left(\frac{\theta}{2}\right)$

$\Rightarrow \cos(\theta) \approx 1 - \frac{\theta^2}{2}$